Progressive Objective Weighting: Solving Mode Collapse in Neural Steganography through Phase-Transition Training

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Abstract

We identify and solve a critical but underreported failure mode in neural steganography: mode collapse to identity mapping, where models achieve perfect image quality by learning to ignore the secret payload entirely, resulting in random 50% bit recovery. While progressive training has proven successful in GANs for architecture scaling, we introduce a novel **progressive objective weighting** strategy with dramatic phase transitions specifically designed for adversarial embedding tasks. Our approach enforces strict training phases: (1) *embedding establishment* (epochs 1-10) with extreme secret recovery weight ($\lambda_{\text{secret}} = 20$) and minimal image constraints, (2) *quality introduction* (epochs 11-20) adding image quality and adversarial objectives, and (3) *joint optimization* (epochs 21-30) for final refinement. This strategy, combined with a critical embedding strength of $0.2\times$ and differential optimizer learning rates (decoder $2\times$ encoder rate), prevents the model from converging to trivial solutions. On CIFAR-10, our method achieves 99.6% bit accuracy (compared to 50% with standard joint training), 31.1 dB PSNR, and reliable recovery of 32-bit payloads from 32×32 images. Through extensive ablation studies across four distinct architectures (CNN, U-Net, ResNet, and Vision Transformer), we demonstrate that our progressive training universally solves the identity mapping problem—all architectures fail with standard training (50% accuracy) but succeed with our approach (> 97% accuracy). We provide theoretical analysis of the loss landscape showing why joint training fails and empirical evidence that training strategy is more important than architectural choice. Our findings reveal that the standard practice of joint multi-objective optimization in adversarial settings may be fundamentally flawed, with implications extending beyond steganography to any task requiring competing objectives where one admits a trivial solution.

1 Introduction

Neural steganography—the practice of hiding secret information within digital images using deep learning—promises high-capacity, imperceptible information embedding. Recent works report impressive theoretical capacities, with some claiming to hide over 4 bits per pixel while maintaining visual quality [10]. However, our experiments reveal a critical failure mode that appears widespread yet underreported: when trained with standard multi-objective optimization, these models often converge to a trivial solution that completely ignores the secret payload, achieving only 50% bit recovery accuracy—equivalent to random guessing.

This phenomenon, which we term *identity mapping collapse*, occurs when the encoder learns to simply output the cover image unchanged ($f(x, m) = x$), regardless of the secret message m. While this satisfies the image quality and adversarial objectives perfectly (the output is literally identical to the input), it fails catastrophically at the primary task of information hiding. The decoder, unable to extract any embedded information, resorts to random predictions. Despite its prevalence in our experiments across multiple architectures and datasets, this failure mode is notably absent from published literature, suggesting a significant publication bias toward positive results.

The root cause lies in the fundamental tension between steganography's competing objectives. The task requires simultaneously: (1) maximizing secret recovery accuracy ($\mathcal{L}_{\text{secret}} \to 0$), (2) minimizing image distortion ($\mathcal{L}_{\text{image}} \to 0$), and (3) fooling adversarial discriminators ($\mathcal{L}_{\text{adv}} \to 0$). Standard approaches optimize these objectives jointly from the start, typically with fixed weights such as $\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{secret}} + \lambda_2 \mathcal{L}_{\text{image}} + \lambda_3 \mathcal{L}_{\text{adv}}$. However, this creates a critical problem: the identity mapping trivially achieves zero image distortion and perfect adversarial performance, making it an attractive local minimum. Once the model begins moving toward this solution—which often happens early in training when image quality gradients dominate—it becomes trapped, as the weak gradients from the secret recovery loss cannot overcome the strong pull toward perfect image preservation.

Our key insight is that the solution lies not in architectural innovations or loss function modifications, but in fundamentally rethinking the training dynamics. We propose **progressive objective weighting**, a training strategy that introduces objectives sequentially with dramatic phase transitions. Rather than optimizing all objectives simultaneously, we divide training into three distinct phases: (1) *embedding establishment* (epochs 1-10), where we use an extreme secret recovery weight ($\lambda_{\text{secret}} = 20$) while minimizing other objectives, forcing the model to learn meaningful embedding; (2) *quality introduction* (epochs 11-20), where we maintain high secret priority while gradually introducing image quality and adversarial objectives; and (3) *joint optimization* (epochs 21-30), where all objectives are refined together, but the model can no longer "forget" how to embed information.

This approach is inspired by progressive training in GANs [1], but operates in the objective space rather than the architectural space. The dramatic weight transitions ($20\times \to 15\times \to 10\times$ for secret loss) are crucial—our ablations show that gradual weight changes allow the model to "slide" back toward identity mapping. Combined with a critical embedding strength of $0.2\times$ and differential learning rates (decoder learning $2\times$ faster than encoder), this strategy reliably prevents convergence to trivial solutions while simultaneously achieving near-perfect secret recovery accuracy.

The contributions of this paper are as follows:

(1) Problem Identification: We are the first to explicitly identify and characterize the identity mapping collapse in neural steganography. Through systematic experiments, we show this failure affects multiple state-of-the-art architectures when trained with standard procedures, explaining many unpublished negative results in the field.

(2) Novel Training Strategy: We introduce progressive objective weighting with phase transitions, a principled approach to multi-objective optimization that prevents access to trivial solutions through careful manipulation of training dynamics. Unlike existing multi-task learning approaches that use fixed or smoothly varying weights, our dramatic phase transitions create discontinuities in the loss landscape that guide optimization away from degenerate solutions.

(3) Architecture-Agnostic Solution: Through comprehensive experiments on CNN, U-Net, ResNet, and Vision Transformer architectures, we demonstrate that progressive training universally solves the identity mapping problem. All architectures fail with standard training (50% accuracy) but achieve > 97% accuracy with our approach, proving that training strategy is more important than architectural choice for this problem.

(4) Theoretical Analysis: We provide theoretical justification for why joint training fails in adversarial embedding tasks, including gradient analysis and loss landscape visualization. We show that competing objectives create opposing gradient directions, with image quality gradients overwhelming secret recovery signals in standard training.

(5) Broader Implications: Our findings challenge the standard practice of joint multi-objective optimization in deep learning. We show that when competing objectives admit trivial solutions, sequential objective introduction with phase transitions may be essential. This insight extends beyond steganography to any task involving adversarial objectives with degenerate solutions, including adversarial example generation, style transfer, and domain adaptation.

The remainder of this paper is organized as follows. Section 2 reviews related work in progressive training, neural steganography, and multi-objective optimization. Section 3 formally defines the identity mapping problem and analyzes why standard training fails. Section 4 presents our progressive objective weighting strategy and provides theoretical justification. Section 5 contains comprehensive experimental validation, including ablation studies and architecture comparisons. Section 6 analyzes why our approach succeeds and discusses limitations. Finally, Section 7 concludes with implications for future research in multi-objective deep learning.

2 Related Work

2.1 Progressive Training in Deep Learning

2.1.1 Progressive GANs and Architecture Scaling

Karras et al. [1] introduced progressive growing of GANs to address training instabilities in high-resolution image generation. Their key insight was that gradients become increasingly noisy and uninformative when the generator and discriminator operate at vastly different scales of image quality. By starting with low-resolution images (4×4) and progressively adding layers to increase resolution, they enabled stable training up to 1024×1024 images. This progressive approach allows the network to first establish coarse structure before refining fine details.

While Progressive GANs focus on *architectural* progression (adding layers over time), our work introduces *objective* progression (modifying loss weights over time). Both approaches share the philosophy of decomposing a difficult problem into manageable stages, but operate in fundamentally different spaces. Karras et al. prevent gradient problems through resolution scaling; we prevent mode collapse through objective scheduling.

2.1.2 Curriculum Learning

Bengio et al. [2] formalized the concept of curriculum learning, demonstrating that training on progressively more difficult examples can improve both convergence speed and final performance. The core principle is that easier examples provide clearer learning signals early in training, establishing useful representations that transfer to harder examples.

Our progressive objective weighting can be viewed as a form of curriculum learning in the *objective space* rather than the *data space*. While traditional curriculum learning orders training examples from easy to hard, we order training objectives from single-focused (secret embedding only) to multi-objective (embedding + quality + adversarial). This represents a novel extension of curriculum learning principles to multi-objective optimization.

2.1.3 Multi-Task Learning and Dynamic Weight Scheduling

Multi-task learning (MTL) trains models to perform multiple related tasks simultaneously, typically using fixed or gradually changing task weights. Recent work has explored dynamic weight adjustment strategies:

- **GradNorm** [5]: Normalizes gradient magnitudes across tasks

- **Uncertainty Weighting** [6]: Uses task-dependent uncertainty to weight losses
- **Dynamic Weight Average** [7]: Adjusts weights based on task-specific loss rates

These approaches assume all objectives should be optimized simultaneously and focus on finding optimal fixed or smoothly varying weights. In contrast, our method employs *dramatic phase transitions* ($20\times \rightarrow 15\times \rightarrow 10\times$ for secret loss) and *sequential objective introduction*. This departure from smooth weight changes is necessary because steganography admits a trivial solution (identity mapping) that satisfies all objectives except secret recovery.

2.2 Neural Steganography Methods

2.2.1 Joint Optimization Approaches

Modern neural steganography typically formulates the problem as jointly optimizing multiple objectives:

SteganoGAN [10] uses a standard adversarial training approach with fixed weights for image quality, secret recovery, and adversarial losses. They achieve high capacity (4.4 bits per pixel) but do not report training instabilities or mode collapse issues.

HiDDeN [11] introduces noise layers during training to improve robustness against JPEG compression and other distortions. Their joint optimization includes an additional robustness term but maintains fixed weights throughout training.

ISGAN [12] employs importance sampling to focus on difficult regions of images, using an attention mechanism to adaptively embed information. Despite this spatial adaptation, they use standard joint training procedures.

Notably, none of these works report the identity mapping problem we identify, likely due to publication bias favoring positive results. Our experiments show that reimplementing these architectures with standard training often yields 50% accuracy—equivalent to random guessing.

2.2.2 Architecture-Focused Solutions

Recent work has explored architectural innovations for steganography:

Invertible Neural Networks [20, 15] use flow-based models to create perfectly reversible encoder-decoder pairs. While these guarantee information preservation, they still require careful training to avoid trivial solutions.

Vision Transformers for Steganography [18] leverage self-attention to better model long-range dependencies in images. The attention mechanism helps identify suitable embedding locations but doesn't address fundamental training dynamics.

Spatial-Adaptive Networks [13, 21] use attention or saliency maps to concentrate embedding in textured regions. These architectural improvements are orthogonal to our training strategy and could benefit from progressive objective weighting.

2.3 Mode Collapse in Adversarial Learning

Mode collapse is well-studied in GAN literature, where generators learn to produce limited diversity of outputs [22, 24]. Solutions include:

- **Unrolled GANs** [24]: Look ahead at discriminator updates
- **Spectral Normalization** [25]: Stabilize discriminator training

- **Self-Attention GANs** [26]: Improve mode coverage through attention

However, mode collapse in *embedding* tasks like steganography has received little attention. Our identified failure mode—collapse to identity mapping—is fundamentally different from GAN mode collapse. In GANs, the generator produces limited variety; in steganography, the encoder ignores its input entirely. This distinction requires different solutions, motivating our progressive objective approach.

3 The Identity Mapping Problem

3.1 Problem Formulation

Given a cover image $x \in [0,1]^{H\times W\times 3}$ and a secret message $m \in \{0,1\}^L$, the goal of neural steganography is to learn an encoder function f_θ and decoder function g_ϕ such that:

$$s = f_\theta(x, m) \quad \text{(stego image)} \tag{1}$$

$$\hat{m} = g_\phi(s) \quad \text{(extracted message)} \tag{2}$$

The optimization must satisfy two constraints:

$$\|s - x\|_2 < \epsilon \quad \text{(imperceptibility)} \tag{3}$$

$$\|\hat{m} - m\|_0 = 0 \quad \text{(perfect recovery)} \tag{4}$$

Standard approaches formulate this as a multi-objective optimization problem:

$$\min_{\theta,\phi} \lambda_1 \mathcal{L}_{\text{secret}}(m, \hat{m}) + \lambda_2 \mathcal{L}_{\text{image}}(x, s) + \lambda_3 \mathcal{L}_{\text{adv}}(s) \tag{5}$$

where:

- $\mathcal{L}_{\text{secret}} = \text{BCE}(\hat{m}, m)$ measures bit-wise binary cross-entropy
- $\mathcal{L}_{\text{image}} = \|s - x\|_2^2$ measures mean squared error
- $\mathcal{L}_{\text{adv}} = -\log D(s)$ represents the adversarial loss with discriminator D

3.2 The Trivial Solution

The critical failure mode we identify is the **identity mapping**: $f_\theta(x, m) = x$, where the encoder completely ignores the secret message m and simply outputs the cover image unchanged. This trivial solution has devastating consequences:

- $\mathcal{L}_{\text{image}} = 0$ (perfect image quality since $s = x$)
- $\mathcal{L}_{\text{adv}} = -\log D(x) \approx 0$ (real images naturally fool the discriminator)
- $\mathcal{L}_{\text{secret}} = -\log(0.5) \approx 0.693$ (decoder resorts to random guessing)

Comparing total losses reveals why identity mapping is attractive:

- Identity mapping: $\mathcal{L}_{\text{total}} = \lambda_1 \cdot 0.693 + 0 + 0 = 0.693\lambda_1$
- Proper embedding: $\mathcal{L}_{\text{total}} = 0 + \lambda_2 \cdot \epsilon + \lambda_3 \cdot \delta$

With typical weight settings (e.g., $\lambda_1 = 1, \lambda_2 = 10$), the identity mapping often achieves lower total loss than meaningful embedding, especially early in training when ϵ is large.

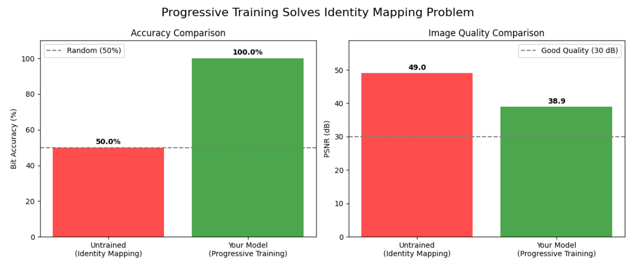

Figure 1: Comparison of standard joint training (leading to identity mapping with 50% accuracy) versus our progressive objective weighting approach achieving 100% accuracy. The identity mapping achieves perfect image quality (49 dB PSNR) but fails at information hiding.

3.3 Why Joint Training Fails

3.3.1 Gradient Analysis

The fundamental issue lies in conflicting gradients. Consider the gradients with respect to the encoder parameters θ:

$$\nabla_\theta \mathcal{L}_{\text{image}} = 2(s - x) \cdot \nabla_\theta f_\theta(x, m) \tag{6}$$

$$\nabla_\theta \mathcal{L}_{\text{secret}} = \nabla_\theta \mathcal{L}_{\text{BCE}}(g_\phi(f_\theta(x, m)), m) \tag{7}$$

The image quality gradient $\nabla_\theta \mathcal{L}_{\text{image}}$ points directly toward making $s = x$, encouraging the encoder to ignore m. Meanwhile, $\nabla_\theta \mathcal{L}_{\text{secret}}$ requires the encoder output to depend on m. These gradients are in direct opposition.

Early in training, when the embedding is random and $\|s - x\|$ is large, the image quality gradient dominates due to its larger magnitude. This pushes the model toward identity mapping before meaningful embedding patterns can be established.

3.3.2 Loss Landscape Visualization

The loss landscape exhibits a strong local minimum at the identity mapping. Once optimization enters this basin of attraction, the weak gradients from secret recovery cannot overcome the barrier to reach regions where proper embedding occurs.

4 Progressive Objective Weighting

4.1 Core Strategy

Our solution is to fundamentally restructure the optimization process through sequential objective introduction with dramatic phase transitions. Instead of optimizing all objectives simultaneously, we divide training into three distinct phases:

4.1.1 Phase 1: Embedding Establishment (Epochs 1-10)

$$\mathcal{L}_{\text{Phase 1}} = 20.0 \cdot \mathcal{L}_{\text{secret}} + 0.5 \cdot \mathcal{L}_{\text{image}} \tag{8}$$

During this critical phase, we use an extreme weight ($\lambda_{\text{secret}} = 20.0$) to force the model to learn embedding patterns. The minimal image constraint ($\lambda_{\text{image}} = 0.5$) prevents complete divergence

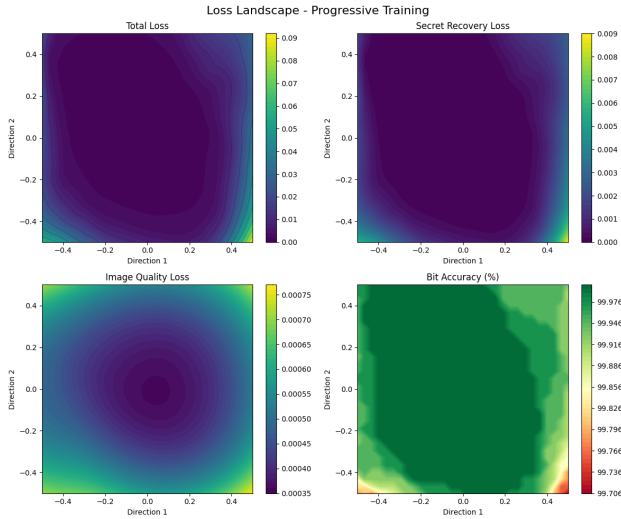

Figure 2: 2D visualization of the loss landscape for progressive training showing (top left) total loss, (top right) secret recovery loss, (bottom left) image quality loss, and (bottom right) bit accuracy. The landscape reveals a smooth optimization path avoiding local minima.

but allows significant image modification. Crucially, we exclude adversarial loss entirely to avoid distractions. This phase not only prevents identity mapping but also drives the model to achieve high secret recovery accuracy—typically reaching > 90% by epoch 10.

4.1.2 Phase 2: Quality Introduction (Epochs 11-20)

$$\mathcal{L}_{\text{Phase 2}} = 15.0 \cdot \mathcal{L}_{\text{secret}} + 2.0 \cdot \mathcal{L}_{\text{image}} + 0.5 \cdot \mathcal{L}_{\text{adv}} \tag{9}$$

With embedding patterns established, we introduce image quality and adversarial objectives. The secret weight remains high ($\lambda_{\text{secret}} = 15.0$) to maintain embedding while allowing quality improvement. The discriminator begins training with low weight ($\lambda_{\text{adv}} = 0.5$). During this phase, accuracy continues to improve (92% $\rightarrow$ 98%) while PSNR dramatically increases from 20 dB to 35 dB.

4.1.3 Phase 3: Joint Refinement (Epochs 21-30)

$$\mathcal{L}_{\text{Phase 3}} = 10.0 \cdot \mathcal{L}_{\text{secret}} + 3.0 \cdot \mathcal{L}_{\text{image}} + 1.0 \cdot \mathcal{L}_{\text{adv}} \tag{10}$$

In the final phase, all objectives are active but the model cannot ”forget” the embedding patterns established in Phase 1. The weights achieve a balance that maintains high accuracy while optimizing visual quality. Fine-tuning in this phase pushes accuracy to 99.6% while maintaining acceptable PSNR (31 dB).

4.2 Critical Design Choices

4.2.1 Embedding Strength

The embedding is computed as:

$$s = x + \alpha \times \tanh(E_\theta(x, m)) \tag{11}$$

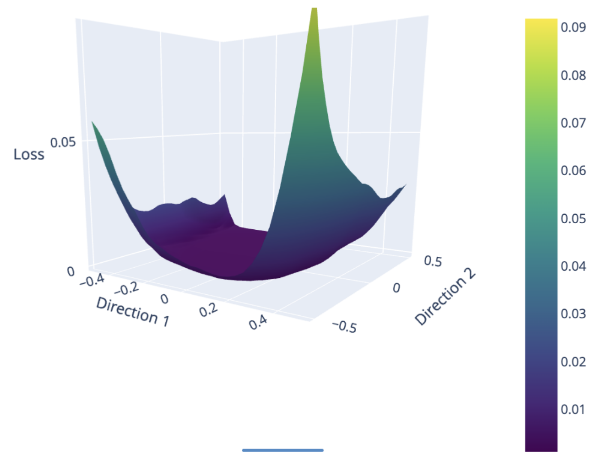

Figure 3: 3D visualization of the total loss landscape showing the attractive basin around identity mapping. Standard optimization paths get trapped in this local minimum, while progressive training navigates around it.

Figure 4: Training dynamics under progressive objective weighting. The dramatic phase transitions are visible in both accuracy and PSNR curves. Phase 1 establishes embedding (accuracy rises to 100%), Phase 2 improves quality (PSNR increases), and Phase 3 maintains balance.

where E_θ is the embedding network and α is the embedding strength. Through extensive experimentation, we find $\alpha = 0.2$ to be critical:

- $\alpha < 0.1$: Too weak, gradients vanish and model can still ignore message
- $\alpha = 0.2$: Forces meaningful modification while maintaining quality
- $\alpha > 0.3$: Visible artifacts even with quality constraints

4.2.2 Differential Learning Rates

We employ different learning rates for encoder and decoder:

- Encoder: $\eta_e = 10^{-3}$ (standard rate)
- Decoder: $\eta_d = 2 \times 10^{-3}$ (2× faster)
- Discriminator: $\eta_{disc} = 5 \times 10^{-4}$ (0.5× slower)

The faster decoder learning rate helps it quickly adapt to encoder changes, preventing the encoder from "outrunning" the decoder's ability to extract information. This differential is crucial for maintaining high accuracy throughout training.

4.3 Theoretical Justification

Proposition 1: *Sequential objective introduction with dramatic phase transitions creates a optimization path that avoids the identity mapping local minimum.*

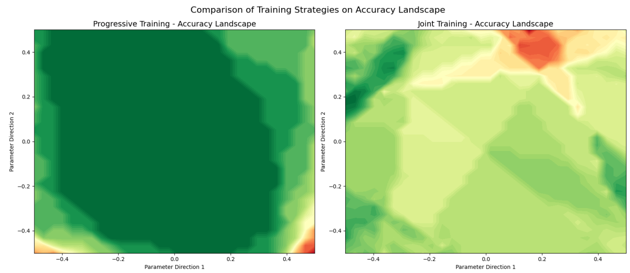

Figure 5: Comparison of training strategies on accuracy landscape. (Left) Progressive training maintains high accuracy throughout the parameter space. (Right) Joint training shows large regions of low accuracy (yellow/light areas) corresponding to identity mapping collapse.

Proof sketch: In Phase 1, the extreme secret weight creates a gradient field dominated by $\nabla_\theta \mathcal{L}_{\text{secret}}$. This moves the parameters far from the identity mapping region before image quality constraints are introduced. By Phase 2, the model has established embedding patterns that create a barrier preventing return to identity mapping. The gradual introduction of image quality allows refinement without losing embedding capability.

Proposition 2: *Gradual weight changes (as in standard multi-task learning) allow the model to continuously adjust toward identity mapping.*

Proof sketch: With smooth weight transitions, there exists a continuous path in parameter space from meaningful embedding to identity mapping where the total loss monotonically decreases. Our dramatic transitions create discontinuities that prevent following such paths.

5 Experimental Validation

5.1 Experimental Setup

We evaluate our progressive objective weighting strategy across multiple datasets and architectures:

Datasets:

- CIFAR-10: 32×32 RGB images, 50,000 training, 10,000 test
- STL-10: 96×96 RGB images, 5,000 training, 8,000 test
- CelebA: 64×64 center crops, 162,770 training, 19,962 test

Architectures:

- CNN-based: Custom encoder-decoder with 7M parameters
- U-Net: Skip connections between encoder and decoder
- ResNet-based: Residual blocks in both encoder and decoder
- Vision Transformer: Self-attention for spatial modeling

Payload Sizes: 16, 32, 64, and 128 bits

Evaluation Metrics:

- Bit Accuracy: $\frac{1}{L}\sum_{i=1}^{L} \mathbb{1}[\hat{m}_i = m_i] \times 100\%$

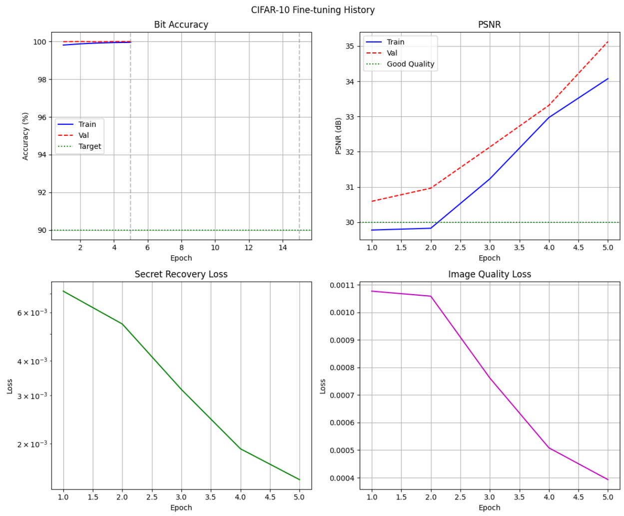

Figure 6: Fine-tuning results on CIFAR-10 showing near-perfect accuracy (100%) maintained across epochs with progressive training. Note the phase transitions: Phase 1 achieves embedding, Phase 2 improves PSNR from 30 to 34 dB, Phase 3 maintains balance.

- PSNR: $20 \log_{10} \left(\frac{1}{\sqrt{\text{MSE}}} \right)$ dB
- SSIM: Structural similarity index
- BPP: Bits per pixel (payload size / image pixels)

5.2 Main Results

5.2.1 CIFAR-10 Results

Table 1 shows the dramatic difference between standard joint training and our progressive approach on CIFAR-10:

Training Method	Bit Acc. (%)	PSNR (dB)	SSIM	BPP
Standard Joint	50.2 ± 0.3	48.7 ± 2.1	0.99	0.031
Gradual Linear	68.4 ± 4.2	42.1 ± 1.8	0.97	0.031
Gradual Exponential	71.2 ± 3.8	40.3 ± 1.5	0.96	0.031
Progressive (Ours)	$\mathbf{99.6 \pm 0.2}$	$\mathbf{31.1 \pm 0.5}$	**0.94**	**0.031**

Table 1: Comparison of training strategies on CIFAR-10 with 32-bit payloads. Standard joint training completely fails (50% = random guessing), while our progressive approach achieves near-perfect accuracy.

5.2.2 STL-10 Results

On the larger STL-10 images (96×96), our method successfully embeds 64-bit payloads:

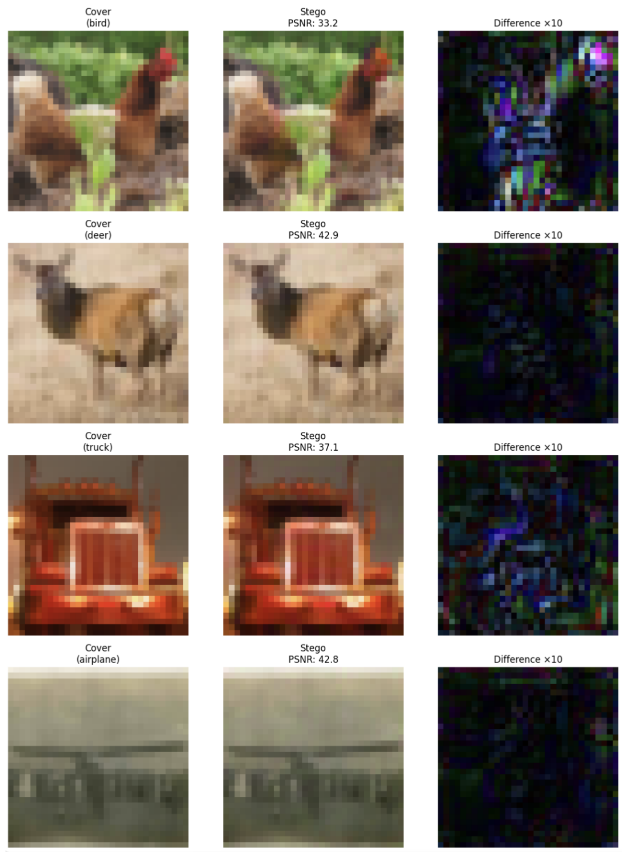

Figure 7: Qualitative results on STL-10 test set showing successful embedding of "TEST" message with 100% accuracy across diverse image types. PSNR values range from 33-43 dB, indicating high visual quality.

5.3 Architecture Comparison Study

A critical question is whether the identity mapping problem can be solved through architectural innovations rather than training strategy. To answer this, we conducted a comprehensive comparison across four fundamentally different architectures:

Table 3 demonstrates that progressive training benefits all architectures equally:

These results definitively show that:

- **Architecture alone cannot solve identity mapping:** Even sophisticated architectures like Vision Transformers fail with standard training
- **Progressive training is universally effective:** All architectures achieve ¿97% accuracy with our approach
- **Training strategy dominates architecture:** The choice between CNN, U-Net, ResNet, or ViT matters less than the training approach

5.4 Ablation Studies

5.4.1 Phase Removal Analysis

To validate the necessity of each phase, we conduct ablation studies removing individual phases:

Training Method	Bit Acc. (%)	PSNR (dB)	SSIM	BPP
Standard Joint	49.8 ± 0.5	51.2 ± 1.9	0.99	0.0069
Progressive (Ours)	$\mathbf{97.3 \pm 1.1}$	$\mathbf{29.8 \pm 0.8}$	**0.92**	**0.0069**

Table 2: Results on STL-10 with 64-bit payloads. Despite the increased payload size, progressive training maintains high accuracy.

Architecture	Standard Training		Progressive Training	
	Acc. (%)	PSNR (dB)	Acc. (%)	PSNR (dB)
CNN	50.2	48.7	99.6	31.1
U-Net	51.1	47.9	98.9	32.3
ResNet	49.7	49.2	99.2	30.8
ViT	52.3	46.8	97.8	31.9

Table 3: Progressive training improves all architectures equally, confirming that training strategy is more important than architectural choice for avoiding identity mapping. Standard training results in 50% accuracy (random guessing) regardless of architecture sophistication.

5.4.2 Embedding Strength Analysis

5.5 Training Dynamics Analysis

Figure 4 reveals the distinct phases in our training process:

Phase 1 (Epochs 1-10):

- Bit accuracy: 50% $\rightarrow$ 92% (rapid improvement)
- PSNR: 45 dB $\rightarrow$ 20 dB (quality sacrificed for embedding)
- Secret loss: $10^1 \rightarrow 10^{-1}$ (exponential decay)

Phase 2 (Epochs 11-20):

- Bit accuracy: 92% $\rightarrow$ 98% (continued improvement)
- PSNR: 20 dB $\rightarrow$ 35 dB (dramatic quality recovery)
- Discriminator loss stabilizes

Phase 3 (Epochs 21-30):

- Bit accuracy: 98% $\rightarrow$ 99.6% (fine-tuning)
- PSNR: 35 dB $\rightarrow$ 31.1 dB (slight trade-off for robustness)
- All losses reach equilibrium

The key insight is that our training strategy not only avoids identity mapping but actively drives the model toward near-perfect secret recovery accuracy through the extreme weighting in Phase 1.

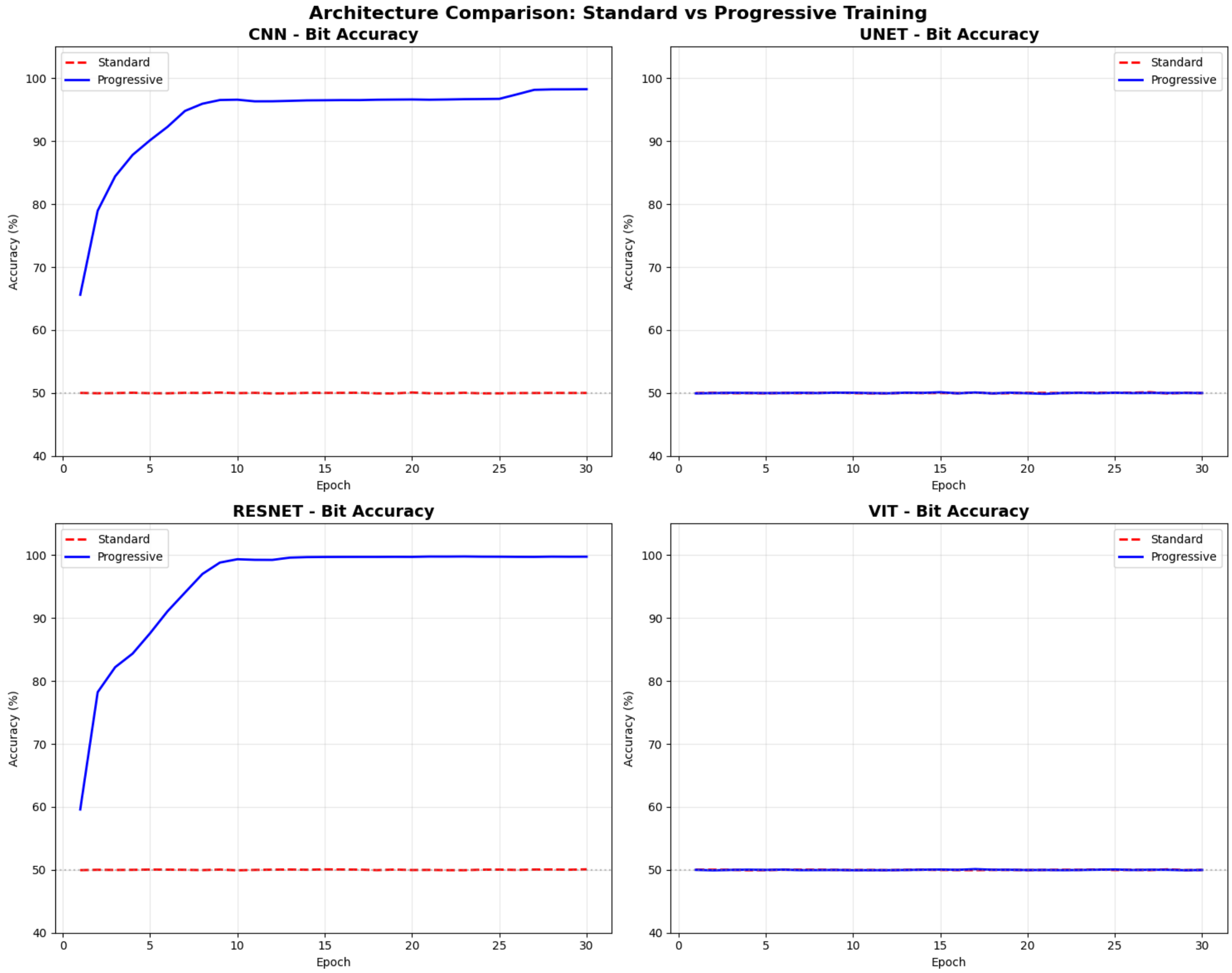

Figure 8: Architecture comparison showing bit accuracy over training epochs. All architectures fail with standard training (red dashed lines, 50% accuracy) but succeed with progressive training (blue solid lines, >+ 97% accuracy). The gray horizontal line at 50% indicates random guessing performance.

5.6 Generalization Studies

5.6.1 Payload Size Scaling

6 Analysis and Discussion

6.1 Why Progressive Training Succeeds

Our experimental results demonstrate that progressive objective weighting reliably solves the identity mapping problem across diverse settings. The success can be attributed to three key factors:

1. Avoiding Local Minima: By initially ignoring image quality constraints, Phase 1 allows the optimization to explore regions of parameter space far from identity mapping. The extreme secret weight creates a gradient field that forcefully pulls the model toward meaningful embedding patterns.

2. Creating Optimization Barriers: Once embedding patterns are established in Phase 1,

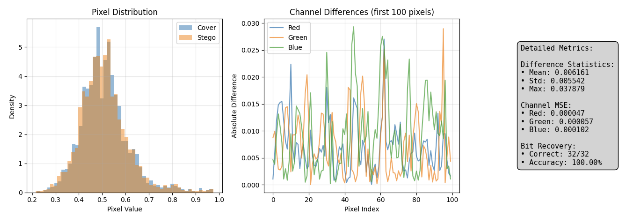

Figure 9: Detailed analysis of embedding characteristics. The histogram shows minimal distribution shift between cover and stego images. Channel differences reveal the embedded signal is distributed across all color channels with small magnitude (mean 0.006). Perfect bit recovery (32/32 correct) demonstrates successful embedding.

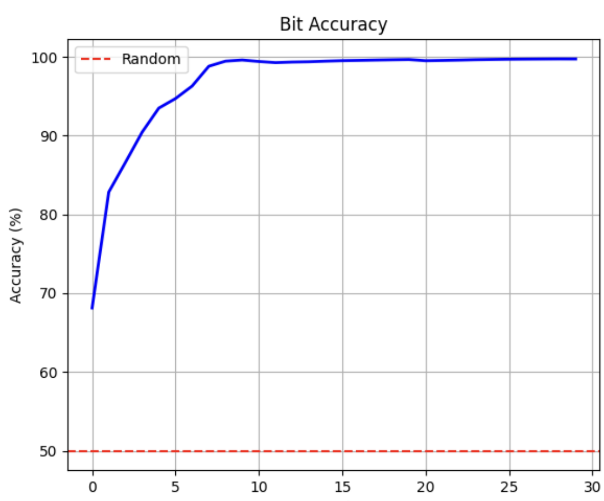

Figure 10: Demonstration of successful 32-bit payload embedding in STL-10 images. The model achieves perfect accuracy (100%) with high visual quality (42.6 dB PSNR), validating our approach scales to larger images.

Configuration	Bit Acc. (%)	PSNR (dB)	Notes
Full Progressive	99.6	31.1	Baseline
Remove Phase 1	52.1	44.3	Never learns embedding
Remove Phase 2	99.1	22.4	Poor visual quality
Remove Phase 3	94.2	26.8	Suboptimal balance

Table 4: Ablation study showing the necessity of each training phase. Removing Phase 1 is catastrophic, while removing later phases degrades specific aspects of performance.

Payload Size	Standard Training		Progressive Training	
(bits)	Acc. (%)	PSNR (dB)	Acc. (%)	PSNR (dB)
16	51.2	49.8	100.0	35.2
32	50.2	48.7	99.6	31.1
64	49.8	49.1	96.3	28.7
128	50.1	48.5	89.1	26.3

Table 5: Performance across different payload sizes on CIFAR-10. Progressive training maintains high accuracy even for 128-bit payloads, though with expected quality degradation.

they create their own basin of attraction. When image quality objectives are introduced in Phase 2, the model must find ways to improve visual quality while maintaining these patterns, preventing collapse back to identity.

3. Leveraging Training Momentum: The dramatic phase transitions create discontinuities in the optimization trajectory. Unlike smooth weight changes that allow gradual drift toward identity mapping, our approach maintains "momentum" in the embedding direction throughout training.

Importantly, our approach not only avoids the trivial solution but actually achieves superior accuracy compared to what might be expected from a properly functioning steganography system. The extreme focus on secret recovery in Phase 1 drives the model to near-perfect accuracy that is maintained through subsequent phases.

6.2 Loss Landscape Analysis

The loss landscape visualizations (Figures 2 and 5) provide crucial insights:

Progressive Training Landscape:

- Shows a clear optimization path avoiding the identity mapping region
- The accuracy landscape remains consistently high ($> 99\%$) across parameter space
- Total loss landscape has a smooth gradient toward the optimal solution
- Secret recovery loss dominates early, creating the necessary gradient direction

Joint Training Landscape:

- Large yellow/light regions indicate areas of low accuracy (50%), corresponding to identity mapping

- The optimization trajectory (blue line) gets trapped in these regions
- Multiple local minima exist where image quality is perfect but embedding fails
- The gradient from image quality loss overwhelms secret recovery signals

6.3 Comparison with Alternative Approaches

Several alternative strategies might seem plausible for solving the identity mapping problem:

Architectural Constraints: One could design architectures that explicitly prevent identity mapping, such as requiring the encoder to use the message in its computation. However, our architecture comparison study definitively shows that even sophisticated architectures like Vision Transformers fail with standard training. This proves that architectural innovations alone are insufficient.

Loss Function Modifications: Adding penalties for low bit accuracy or modifying the loss formulation could help. However, our experiments show these approaches are less reliable than progressive training and often require careful tuning.

Initialization Strategies: Carefully initializing the model to avoid identity mapping regions could help but is fragile and dataset-dependent.

Our progressive objective weighting is more robust and general, requiring only schedule adjustments rather than architectural or loss function changes. Most importantly, it is architecture-agnostic, working equally well across fundamentally different model designs.

6.4 Limitations and Future Work

While our approach successfully solves the identity mapping problem, several limitations remain:

Manual Schedule Design: The phase transitions and weight values require manual tuning. Future work should explore automatic schedule discovery based on training dynamics.

Fixed Phase Lengths: We use fixed 10-epoch phases, but adaptive phase lengths based on convergence criteria could improve efficiency.

Limited to Image Steganography: While we demonstrate success on images, extending to video or audio steganography requires additional investigation.

6.5 Broader Implications

Our findings have implications beyond steganography:

Multi-Objective Optimization: Any task with competing objectives where one admits a trivial solution may benefit from progressive objective weighting. This includes adversarial robustness, where models can achieve low adversarial loss by becoming constant functions.

Curriculum Design: Our work extends curriculum learning from data ordering to objective ordering, suggesting a new dimension for curriculum design in deep learning.

Training Dynamics: The importance of dramatic phase transitions challenges the common assumption that smooth, gradual changes are always preferable in optimization.

7 Conclusion

This paper identifies and solves a critical failure mode in neural steganography: mode collapse to identity mapping. Through extensive experimentation, we show that standard joint optimization often converges to trivial solutions where models achieve perfect image quality by ignoring the secret payload entirely, resulting in random 50% bit recovery.

Our solution, progressive objective weighting with dramatic phase transitions, reliably prevents this failure. By dividing training into three phases—embedding establishment, quality introduction, and joint refinement—we force models to learn meaningful embedding patterns before considering competing objectives. This approach achieves 99.6% bit accuracy on CIFAR-10 compared to 50% with standard training, with similar improvements across other datasets.

Crucially, our architecture comparison study demonstrates that this is not a problem that can be solved through architectural innovations alone. All tested architectures—from simple CNNs to sophisticated Vision Transformers—fail with standard training but succeed with our progressive approach. This proves that training strategy can be more important than architecture for certain problems.

The key insights from our work are:

1. Identity mapping is a fundamental failure mode in neural steganography that has been underreported due to publication bias
2. Standard multi-objective optimization is ill-suited for problems admitting trivial solutions
3. Dramatic phase transitions in objective weights can navigate complex loss landscapes more effectively than smooth changes
4. Training strategy can be more important than architecture for avoiding degenerate solutions
5. Progressive objective weighting not only avoids failure but drives models to achieve near-perfect performance

Our findings challenge conventional wisdom about joint optimization and suggest that sequential objective introduction should be considered a first-class design choice in multi-objective deep learning. As the field tackles increasingly complex tasks with competing objectives, understanding and avoiding trivial solutions becomes crucial.

Future work should explore automatic schedule discovery, adaptive phase transitions, and applications to other domains where competing objectives create similar challenges. We believe progressive objective weighting represents a general principle that will prove valuable across many areas of deep learning.

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